

What a machine reads off your lips

Every line below was produced from video alone — no audio, at any stage. The model watched a mouth move and guessed at the words. Here is how close it got, measured across the full 1,321-clip benchmark.

34.28%

of words wrong,
overall

415

of 1,321 clips read
exactly

10×

faster than real time

50 min

of video scored

Said, and read

The top word in each pair is what the speaker actually said. The bottom word is what VoxLens read off their lips.

 matched  wrong word  invented or missed

READ EXACTLY

0% word errors · 3.8s

the choices don't make sense because it's the wrong question
the choices don't make sense because it's the wrong question

READ EXACTLY

0% word errors · 1.4s

you made it work for me
you made it work for me

CLOSE, BUT WRONG

22% word errors · 3.0s

for depression it was  four  — and a half times

for depression it was a few and a half times

CLOSE, BUT WRONG

22% word errors · 2.0s

so i'd like to leave you with these thoughts
so i'd like to leave you with this thought

CLOSE, BUT WRONG

22% word errors · 3.1s

— — we have problems that we desperately need to solve
i mean we have problems that we desperately need to solve

LOST THE THREAD

167% word errors · 2.2s

and in fact it probably won't — — — —
it's the same thing right we're on the running water

LOST THE THREAD

180% word errors · 1.8s

you're doing incredibly important work — — — —
so i'm going to talk about my first one

Why it confuses those particular words

Look at what it swaps. *These* becomes *this*. *Thoughts* becomes *thought*. These are not random errors — they are words that look almost identical on the mouth.

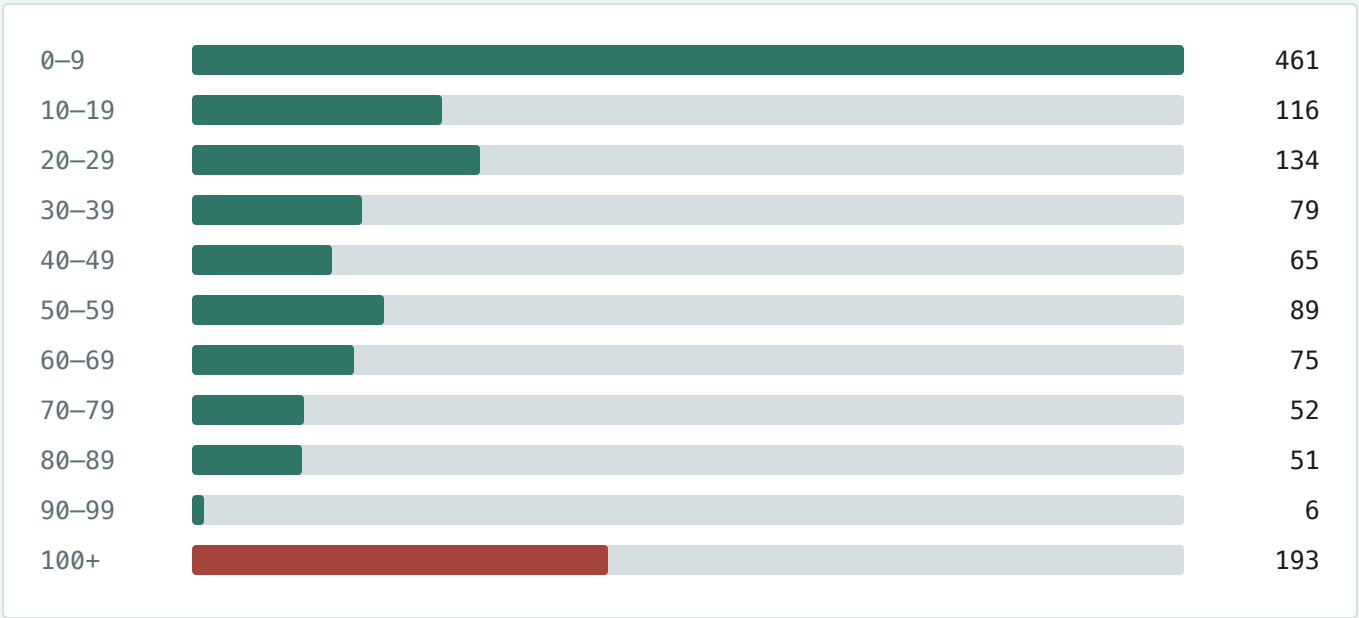
p · b · m → lips together, then apart
“pat”, “bat” and “mat” are the same video.

The difference between them is voicing and airflow, both of which happen inside the throat where no camera can see. English has around 44 distinct sounds but only about a dozen distinguishable mouth shapes, so entire groups of words collapse into one

another. Human lip-readers hit roughly 30–45% word accuracy on unconstrained speech for the same reason. Context has to do the rest of the work — for the machine as much as for a person.

It rarely half-works

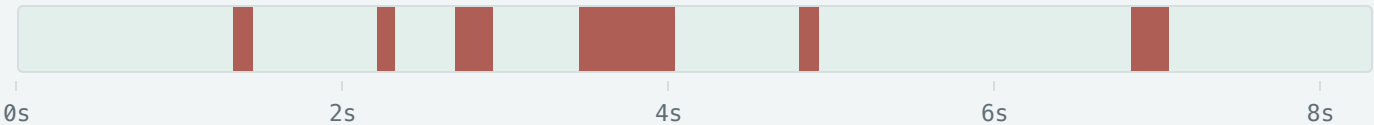
The failure pattern is not a gentle slope. Clips cluster at the ends: read perfectly, or lost entirely. 461 clips came in under 10% word errors, while 193 were worse than 100% — the model inventing more words than were spoken.



Clips by word error rate, all 1,321 in the benchmark.

When it cannot see the mouth

On real footage the face turns away, blurs, or leaves frame. VoxLens marks those stretches rather than quietly inventing words to fill them. This is one 8.3-second clip; the marked bands are the moments it was reading nothing.



6 unreadable stretches, 61 of 208 frames. The longest ran 0.6 seconds — long enough to lose several words. The tool reports where they fall, and stops short of claiming which

words they cost, because that would mean pretending to a precision it does not have.

What this is measured on. The LRS3 benchmark test split, 50 minutes of video, 9,890 words, greedy decoding. Benchmark video is cleanly shot and cleanly cropped. On in-the-wild footage the same model scores closer to 48% word errors — roughly every other word — so treat the number above as the optimistic end.

These figures are a regression baseline for this pipeline measured against itself, not a claim of parity with published research.